

MAD Lab #3

Submitted By

Student Name: Abdul Rehman

Registration No: 04072312041

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BS Computer Science QAU



Output:

```
PS C:\Users\Administrator\Desktop\MAD\Lab 3> dart run --enable-asserts lab3.dart
Seed: 41 (t=4, u=1)
--- Step 1 ---
--- Step 2 ---
--- Step 3 ---
--- Step 4 ---
--- Step 5 ---
--- Step 6 ---
--- Step 7 ---
--- Step 8 ---
--- Step 9 ---
--- Step 10 ---
PS C:\Users\Administrator\Desktop\MAD\Lab 3>
```

Activate Windows
Go to Settings to activate Windows.

Ln 20, Col 12 Spaces: 2 UTF-8 CRLF Dart Finish Setup Claude Code 11:25 PM 10/4/2026

Step 1 Output:

```
PS C:\Users\Administrator\Desktop\MAD\Lab 3> dart run --enable-asserts lab3.dart
--- Step 1 ---
Step 1: Latte Rs 119
Step 1: Mocha Rs 125
PS C:\Users\Administrator\Desktop\MAD\Lab 3>
```

Ln 66, C

Step 2 Output:

```
PS C:\Users\Administrator\Desktop\MAD\Lab 3> dart run --enable-asserts lab3.dart
--- Step 2 ---
Step 2: Latte Rs 119
Step 2: Test Special Rs 80
PS C:\Users\Administrator\Desktop\MAD\Lab 3>
```

Ln 74, Co

Step 3 Output:

```
PS C:\Users\Administrator\Desktop\MAD\Lab 3> dart run --enable-asserts lab3.dart
--- Step 3 ---
Step 3: Water Rs 0
Step 3: Samosa Rs 133
Step 3: floor=80, free price=0
PS C:\Users\Administrator\Desktop\MAD\Lab 3>
```

Ln 60, Col 11

Step 4 Output:

```
PS C:\Users\Administrator\Desktop\MAD\Lab 3> dart run --enable-asserts lab3.dart
• --- Step 4 ---
Step 4: same object? true
Step 4: entries = 3
Step 4: last = order #403
○ PS C:\Users\Administrator\Desktop\MAD\Lab 3>
Ln 72, Col 11
```

Step 5 Output:

```
Step 4: last = order #403
PS C:\Users\Administrator\Desktop\MAD\Lab 3> dart run --enable-asserts lab3.dart
• --- Step 5 ---
Step 5: Latte x2
Step 5: total=238 tax=21
Step 5: assert fired
PS C:\Users\Administrator\Desktop\MAD\Lab 3>
Ln 98, Col 11 Sp
```

Step 6 Output:

```
• PS C:\Users\Administrator\Desktop\MAD\Lab 3> dart run --enable-asserts lab3.dart
--- Step 6 ---
Step 6: grand=259
Step 6: big order? false (limit 530)
Step 6: label=Latte x2
○ PS C:\Users\Administrator\Desktop\MAD\Lab 3>
Ln 1, Col 1 S
```

Step 7 Output:

```
Step 6: label=Latte x2
• PS C:\Users\Administrator\Desktop\MAD\Lab 3> dart run --enable-asserts lab3.dart
--- Step 7 ---
Step 7: topped up -> 460
Step 7: bad value -> 0
Step 7: reset -> 679
Step 7: paid order -> 420
○ PS C:\Users\Administrator\Desktop\MAD\Lab 3>
Ln 188, Co
```

Step 8 Output:

```
PS C:\Users\Administrator\Desktop\MAD\Lab 3> dart run --enable-asserts lab3.dart
• --- Step 8 ---
Step 8: menu = [Latte (Rs 119), Brownie (Rs 140), Fries (Rs 161), Chai (Rs 112)]
Step 8: priciest = Fries
Step 8: sum = 532
PS C:\Users\Administrator\Desktop\MAD\Lab 3> 
```

Activating Extensions... Ln 189, Co



Step 9 Output:

```
PS C:\Users\Administrator\Desktop\MAD\Lab 3> dart run --enable-asserts lab3.dart
• --- Step 9 ---
Step 9: Latte x1 = 129
Step 9: Brownie x2 = 305
Step 9: Fries x3 = 526
Step 9: receipt total = 960
Step 9: log size = 3
PS C:\Users\Administrator\Desktop\MAD\Lab 3> 
```

Activating Extensions... Ln 190, Col 11 Spac



Step 10 Output:

```
PS C:\Users\Administrator\Desktop\MAD\Lab 3> dart run --enable-asserts lab3.dart
• --- Step 10 ---
Step 10: CAFE41 gives 10% off, min spend 700
Step 10: cached? true
Step 10: receipt 960, discount 96, payable 864
PS C:\Users\Administrator\Desktop\MAD\Lab 3> 
```

Activating Extensions... Ln 191, Col 12



Q1. `Animal(this.name, this.type);` and the verbose constructor give the same result. What does the shorthand save you?

The shorthand saves us from writing the field assignments manually. It automatically assigns the constructor parameters to the fields.

Q2. When would you choose a named constructor, and when a factory constructor?

A named constructor is useful when a class has different ways of initializing an object. A factory constructor is useful when object creation needs special control, such as returning a cached object.

Q3. What is the difference between assigning a field in a constructor body and assigning it in an initializer list?

An initializer list runs before the constructor body and is required for initializing final fields. The constructor body runs afterward.

Q4. Give one reason to use a getter instead of storing the value in a field, and one reason to use a setter instead of a public field.

A getter can calculate a value when it is requested instead of storing the calculated value. A setter can validate or modify a value before storing it.